DOM PROJECT — ANNEX

Anex 1 - Performance Testing Plan - Peak Month Capacity Simulation

Executive recommendation to extend the DOM performance testing strategy with peak-month capacity validation.

Draft Update | Average Volume: ~20 TB/month | Peak Volume: ~65 TB/month | Approach: Simulation + Extrapolation

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Purpose

This note summarizes a recommended update to the DOM performance testing strategy based on the latest ingestion volume estimate.

Important: This is not a final performance report. It is a proposed adjustment to the test approach for management and client alignment.

Key Update

Average Monthly Volume

~20 TB

Estimated average monthly ingestion volume.

Historical Peak Month

~65 TB

Estimated historical peak monthly ingestion volume.

Peak Daily Equivalent

~2.17 TB/day

Assuming a 30-day month.

This means the performance plan should validate not only individual pipeline runs, but also whether the platform can support expected monthly and peak-month operational demand.

Current Plan Remains Valid

The existing performance scenarios should remain in scope because they validate job-level performance:

- **500 GB / 10M rows in less than 4 hours**
- **1 TB / 20M rows scalability test**
- Large file scenario, such as a **50 GB CSV**
- High file-count scenario, such as **50,000 small CSV files**
- Data quality and quarantine overhead
- Concurrent pipeline execution

> These tests are necessary, but they do not fully answer whether the platform can support a peak month of approximately **65 TB**.

Recommended Addition

Peak Month Capacity Simulation — 65 TB

Objective: Estimate whether DOM can support a historical peak ingestion month of approximately **65 TB**, within acceptable processing time, infrastructure capacity, and cost.

Recommended Approach

Do **not** start by processing 65 TB end-to-end. Instead, use a phased simulation model:

1. **Run Benchmarks** — Execute representative 500 GB and 1 TB pipeline runs.
2. **Measure Throughput** — Capture actual GB/hour and runtime.
3. **Measure Cost** — Capture DBU consumption and estimate cost/TB.
4. **Sustained Window** — Run 4–8 hour sustained test if approved.
5. **Concurrency** — Run 2 and 4 simultaneous pipelines.
6. **Extrapolate** — Estimate time and cost for 65 TB.

Why This Approach

A full 65 TB execution may be expensive, time-consuming, and difficult to clean up. A phased approach provides management-level confidence while controlling cost and risk.

It will help determine:

- Whether the current platform can support peak-month demand.
- Whether concurrency is required.
- How long 65 TB would take to process.
- Whether the expected processing window is realistic.
- What the estimated cost per TB and peak-month cost would be.
- Whether Databricks, Spark, Storage, Unity Catalog, or job scheduling become bottlenecks.

Required Management / Client Decisions

1. Is **65 TB/month** an official planning target for peak capacity?
2. What is the expected processing window: continuous, daily batch, overnight window, or project-based processing?
3. Is extrapolation acceptable, or is a larger direct execution required?
4. What concurrency level should be considered realistic for production?
5. What cost guardrails should be applied during testing?

Recommended Next Steps

1. Add **Peak Month Capacity Simulation** to the performance test plan.
2. Confirm the 20 TB average and 65 TB peak assumptions with stakeholders.
3. Execute baseline 500 GB and 1 TB tests.
4. Capture throughput, duration, DBU usage, and cost per TB.
5. Run limited sustained and concurrency tests.
6. Estimate required processing time and cost for 65 TB.
7. Present findings and recommendations before approving any full-scale test.

Summary Recommendation

The performance strategy should be expanded from job-level testing to include **monthly capacity validation**.

> The recommended approach is to simulate the 65 TB peak month using measured results from representative workloads, sustained execution, and limited concurrency, rather than immediately executing a full 65 TB test.

This provides a practical balance between confidence, cost control, and production-readiness planning.